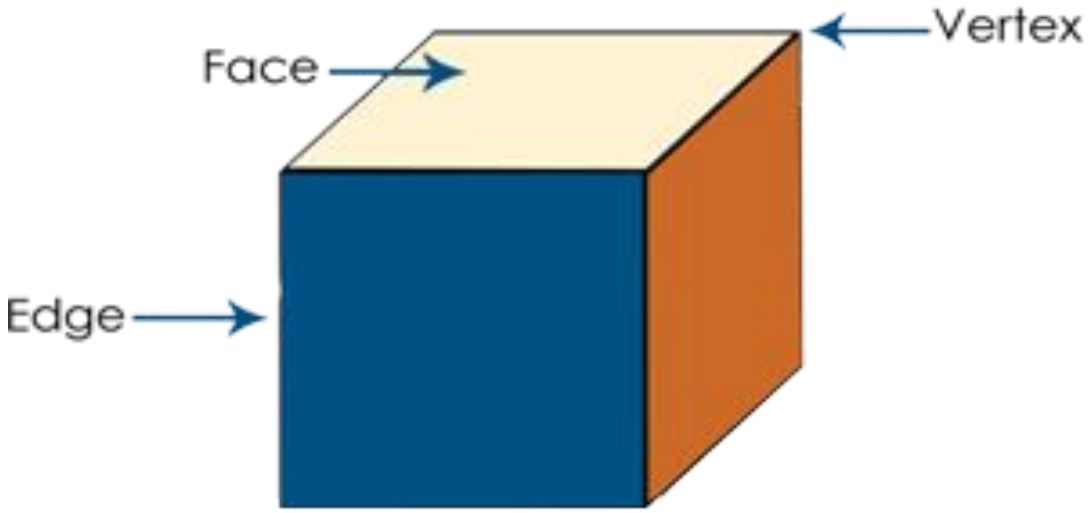

FACE

Cubes

FACE

Cubes



FACE

Cuts and Pieces

N cuts will give N+1 pieces. If we have L number of cuts, B number of cuts, H number of cuts along length, breadth and height of a cube respectively, then Total number of pieces will be, (L+1)(B+1)(H+1).

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Question 1:

Directions for questions 1 to 3: A solid cube of side 8 cm has been painted red, blue and black on pairs of opposite faces. It is then cut into cubical blocks of side 2 cm each.

1. How many cubes have no face painted?

a. 1

b. 4

c. 8

d. 27

FACE

Question 2:

How many cubes have only 1 face painted?

a. 18
b. 22
c. 24
d. 8

Question 3:

3. How many cubes have only 2 faces painted?

a. 8
b. 16
c. 20
d. 24

Question 4:

Directions for questions 4 to 8:

A solid cube is painted red on two adjacent sides and black on the sides opposite to the red sides, and green on the remaining sides. It is then cut into 64 smaller cubes of equal size.

4. How many cubes have only one side coloured?
a. 16
b. 24
c. 28
d. 32

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Question 5:

5. How many smaller cubes have at least one side as red?

a. 16
b. 24
c. 28
d. 32

Question 6:

6. How many cubes are there which are red on one side and black on the opposite side?

a. 0
b. 4
c. 8
d. 16

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Question 7:

7. How many cubes have three sides painted?

a. 24
b. 8
c. 16
d. 20

Question 8:

8. How many cubes have one face green and one of the adjacent faces black or red?

a. 8
b. 16
c. 24
d. 28

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Question 9:

Directions for questions 9 to 12:
A cube has the following figures drawn on five of its faces: circle, cross, triangle, square and ellipse. The top surface is blank. The ellipse is between the cross and the triangle and the square is on the triangle's right. The ellipse and the square are opposite to each other.
Answer the following questions based on the given information.

9. Where is the circle?

a. Left of the triangle
b. On top
c. Bottom face
d. Opposite the triangle

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Question 10:

10. How many arrangements are possible with the given data?

a. Only one

b. Two

c. Three

d. Four

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Question 11:

11. If the cube is rotated such that the face having the ellipse is towards north, then which figure will face towards west?

a. Square
b. Triangle
c. Circle
d. Cross

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Question 12:

12. The cube is rolled to get the ellipse on the top face. What is on the bottom face of the cube?

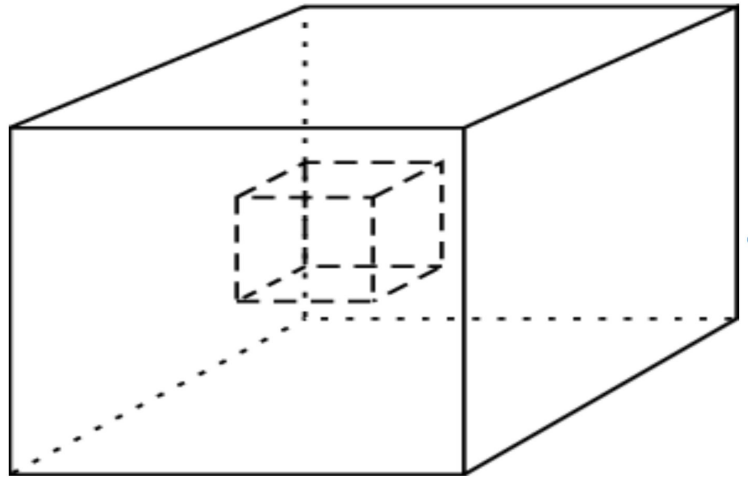
a. Square
b. Circle
c. Blank face of the cube
d. Triangle

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Question 13:

13. A hollow cube of size 5 cm is taken. It is made of smaller cubes of size 1 cm. If 4 faces of the outer surface of the cube are painted, totally how many faces of the smaller cubes remain unpainted?

a. 800
b. 488
c. 900
d. 500



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Question 14:

Directions for questions 14 to 16:A solid cube of side 9 cm has been painted green, red and yellow on pairs of opposite faces. It is then cut into cubical blocks of each side 3 cm.

14. How many cubes have none of the faces painted?

a. 1

b. 6

c. 8

d. 4

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Question 15:

15. How many cubes have three faces painted?

a. 2
b. 4
c. 8
d. 12

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Question 16:

16. How many cubes have four faces painted?

a. 0
b. 1
c. 2
d. 3

Question 17:

17. What is the least number of cuts required to cut a cube into 84 identical pieces, assuming all cuts are made parallel to the faces of the cube?

a) 11
b) 12
c) 13
d) 10

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Question 18:

18. A cube having an edge of 12 cm each. It is painted red on two opposite faces, blue on one other pair of opposite faces, black on one more face and one face is left unpainted. Then it is cut into smaller cubes of 1 cm each. Find the no. of smaller cubes which are having 0-face painted.

a) 1100
b) 1000
c) 900
d) 800

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Question 19:

19. A cube of side 20cm is of blue colour with a 4cm wide yellow strip along all the sides on all the faces. The cube is then cut into 125 smaller equal size cubes. How many cubes have at least two yellow faces each?

a) 125
b) 44
c) 54
d) 27

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Question 20:

20. One hundred and twenty-five cubes of the same size are arranged in the form of a cube on a table. Then a column of five cubes is removed from each of the four corners. All the exposed faces of the rest of the solid (except the face touching the table) are coloured red. How many cubes do not have any coloured face?

a) 125
b) 44
c) 54
d) 36

FACE

THANK YOU

FACE